

TABLE III
COMPARISON BETWEEN DIFFERENT TYPES OF SOLAR CELLS

Bulk type/wafer based (crystalline)						
Type	Mono-crystalline	Poly-crystalline	Thin-film type			
			Amorphous Si	CdTe	CIGS	Polymer organic
Area (in m²) required for 1kW	6-9	8-9	13-20	11-13	9-11	—
Typical length of warranty	25 years	25 years	10-25 years			
Temperature resistance	Performance drops 10 to 15 per cent at high temperature	Less temperature resistant than monocrystalline	Tolerates extreme heat	Relatively low impact on performance		—
Additional details	Oldest cell technology and most widely used	Less silicon waste in the production process	• Tend to degrade faster than crystalline-based solar panels • Low availability on the market			
Pros	High efficiency	High efficiency with respect to price	Low price	• Low price • Able to automate all manufacturing processes		• Less manufacturing • Can be more efficient (still under research)
Cons	Increase in manufacturing cost caused by the supply shortage of silicon		Low efficiency			

(Courtesy: www.solarlove.org)

effective as compared to other renewable power sources.

- It requires less capital investment and low maintenance.

Disadvantages of solar cells

However, there are also some disadvantages such as:

- Solar cells cannot be used in the absence of the Sun.
- Power generated during cloud cover is less.
- A large geographical area is required to deploy solar panels.

Applications

Solar cells are extensively used in calculators and watches. Many people arrange solar panels on the rooftops of their homes or in open areas to generate electric power to operate various devices.

Solar cells are also widely used in Space applications. Researchers have developed multi-junction solar cells especially for spacecraft power systems to increase their power efficiency. Spectrolab, a California based company, is the world's leading manufacturer of high-efficiency multi-junction Space solar cells and panels, and currently provides power to hundreds of satellites and interplanetary spacecrafts.



Fig. 6: Solar panels

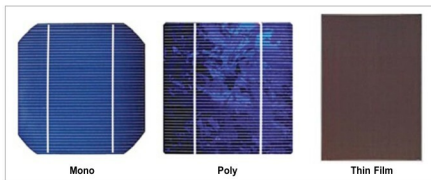


Fig. 7: Types of solar cells

Solar cells are connected in series and parallel to form solar panels that provide more voltage and current. Figs 6 and 7 show solar panels

and different types of solar cells, respectively. Table III shows the pros and cons of the different types of solar cells. **EFY**